

## **Evolutionary Perspectives on Ontology and Epistemology**

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**Abstract:** In recent years, more and more cognitive scientists and philosophers of mind have invoked evolution in explaining both the existence and content of mental states. In this paper, I spell out some of the ontological and epistemological consequences of explaining the mind in terms of evolution by natural selection. In particular, I argue that what I call strongly evolutionary theories of mind do not sit happily with objectivism. I end by sketching some consequences for the methodology of cognitive science.

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### **0. Introduction: The Pyramid of Being**

In a recent paper, Tim van Gelder (1998) helpfully categorizes all the recent attempted solutions to the mind-body problem according to their answers to just five questions: Is the mental *identical* to the physical? Is the mental *reducible* to the physical? Is the mental physically *realized* or *constituted*? Does the mental *supervene* on the physical? Do the mental and physical *interact causally*? van Gelder’s purpose in characterizing the attempted solutions via these five questions is to show the background assumptions that all the participants in the debate share. van Gelder points to four such background assumptions, and rejects them all, on the way to describing a fundamentally new position on the relation between the mental and the physical. But there is at least one background assumption that van Gelder does not point out (presumably because he does not disagree with it): the assumption that it is the relation between the mental and the *physical* that is at issue. In other words, van Gelder assumes, like everyone else in the debate, that the world to which the mind is related is physical. In this paper, I will

argue that that assumption is unwarranted.

To start, I will say a few things about what the common assumption I will be rejecting comes to.

Dennett, who is typical of participants in the debate in this regard, pledges his allegiance to what he calls “Standard Scientific Epistemology and Metaphysics” (1993, p.234) and says that

The prevailing wisdom, variously expressed and argued for, is *materialism*: there is only one kind of stuff, namely *matter*--the physical stuff of physics, chemistry, and physiology--and the mind is somehow nothing but a physical phenomenon. In short, the mind is the brain. According to the materialists, we can (in principle!) account for every mental phenomenon using the same physical principles, laws and raw materials that suffice to explain radioactivity, continental drift, photosynthesis, reproduction, nutrition and growth. (1991, p.33)

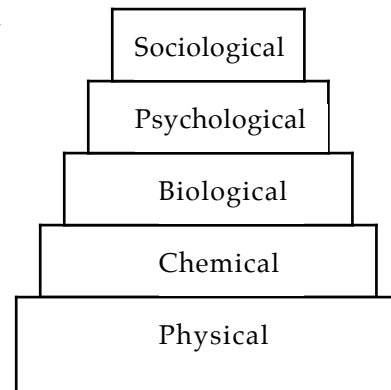


Figure 1

Once we agree to this much, the rest is only detail work, spelling out more precisely how it is that mental stuff is physical stuff.

The usual answers involve what we might call the “Pyramid of Being”. The Pyramid of Being is a conception of nature (and scientific theories) as divided into more basic or more general levels, like the physical or quantum mechanical, which come at the bottom of the pyramid, and the less basic or less general levels, like the psychological or sociological, sitting on top of those. We can see the usual conception in figure 1. We might fill this in a bit, putting ‘Biochemical’ and ‘Physiological’ and ‘Superstrings’ in the appropriate places, but this restricted version will do for now.

The relation between these levels is the main source of controversy among adherents to Standard Scientific Epistemology and Metaphysics. There are two main positions one might take. The

first, rather old-fashioned, position is the one van Gelder calls “baseline physicalism”. In baseline physicalism, every entity at a higher level is identical to some entity or entities at a lower level, all the way down to the quantum mechanical. Furthermore, the relation between theories is (eventual, in principle) *reduction*. The main alternate position, non-reductive physicalism, denies that higher-level theories can be reduced to lower level ones.

Correspondingly, non-reductive physicalists claim that higher level entities are not identical to lower-level entities, or collections thereof. They claim, instead, that the higher level entities *supervene* on lower level entities; or, equivalently, the lower level entities *implement* the higher level ones.<sup>i</sup> For the non-reductive physicalist, the difference between higher and lower-level theories and entities is their *generality*; everything is physical and can be described by the laws of physics, comparatively fewer things are chemical, and so on up the pyramid. Although a great deal of ink has been spilled over which of these conceptions of nature (baseline or non-reductive physicalism) is the correct one, there is the broad agreement that the world is *at bottom* physical.

In what follows, I will argue that currently popular, teleological theories of cognition (e.g. that of Millikan 1984), which I will call *strongly evolutionary theories*, are incompatible

with the Pyramid of Being, because they make it extremely unlikely that the world to which minds are related is *at bottom* physical. Thus we will have a dilemma: either strongly evolutionary theories of cognition are false or nature is not like the Pyramid of Being. I will not take this incompatibility as a *reductio* of the theories of mind;<sup>ii</sup> instead, I will take it to indicate that the Pyramid of Being is not an accurate depiction of nature. In so doing, I will not just be arguing that strongly reductionistic physicalism (what van Gelder calls “baseline physicalism”), according to which everything that is identical with and reducible to some set of physical things, is false. This has been done many, many times. I will also not be arguing for any sort of dualism or other rejection of naturalism: the conclusions I reach will be thoroughly naturalistic.<sup>iii</sup> Instead, I will argue that *all* of the standard positions on the mind body problem (baseline physicalism, non-reductive physicalism, and various dualisms) are false. This is the case, I will argue, because they all assume that the world to which the mind is related is the world as described by physics. Along the way, I will spell out what I take to be some of the consequences of taking this horn of the dilemma for the philosophy of mind and cognitive science.

## 1. Strongly Evolutionary Theories of Cognition

Call a theory of cognition *strongly evolutionary* just in case it ascribes content to states of animals in virtue of the evolutionary history of those animals. In strongly evolutionary theories, the fact that states of an animal have content is explained by referring to the biological function of those states; the function is, in turn, determined, by the role of the states (or the ability to produce and use those states) in the animal's evolutionary history. Some examples of strongly evolutionary theories of cognition are: Millikan's teleosemantics (1984, 1993); Papineau's teleological theory of content (1987); Gibson's ecological psychology (Gibson 1979, Michaels and Carello 1981); Dennett's theory of sub-personal content in *Content and Consciousness* (1969); and Andy Clark's more recent exposition of the theory of "embodied, active cognition" (Clark 1997).<sup>iv</sup>

In strongly evolutionary theories of cognition, it is generally agreed that for a state to impact an animal's chances of survival, it must lead to changes in the behavior of the animal. That is, any state of an animal that is to have an effect on an animal's chances for survival and reproduction--which it must to have content--must be suitable for guiding behavior. For example, it does no evolutionary good to know that there is water over the hill if that knowledge doesn't help you to actually find and drink the water. So in animals that do not have the ability to make complicated inferences, there will be simple contentful states that lead directly to behavior. Thus Gibsonians talk of *affordances*, environmental structures that are sensed by an animal directly as opportunities for behavior; Millikan (1996) speaks of what she calls *pushmi-pullyu* representations, after the two-headed llama from Doctor Dolittle; and Clark (1997) speaks of *action-oriented* representations. The Gibsonians, Millikan and Clark all agree that these structures (affordances, pushmi-pullyu and action-oriented representations) will be evolutionarily more basic than their purely descriptive or directive cousins, beliefs and desires. As Millikan writes:

PPRs [pushmi-pullyu representations] have both a descriptive and a directive function, yet they are not equivalent to the mere conjunction of a pure descriptive representation and a pure directive one but are more primitive than either. Purely descriptive and purely

directive representations are forms requiring a more sophisticated cognitive apparatus to employ them than is necessary for these primitives. Purely descriptive representations must be combined with directive representations through a process of practical inference in order to be used by the cognitive systems. Purely directive representations must likewise be combined with descriptive ones. The employment of PPRs is a much simpler affair. (1996, p.185)

Furthermore, such basic states will not represent things in an activity-neutral way. These states will tend to stand for only those features of the environment that are relevant for guiding action, those features that are significant to the animal and its projects. Clark, for example, claims that his action-oriented representations are both local and personal: they are local in that they relate to the circumstances currently surrounding the agent; they are personal in that they are related to the agent's needs and the skills that it has. Since action-oriented representations are attached to a particular agent and its situation, they will represent the world as a group of opportunities for action. Put most simply, such states stand for the relevant things happening to the agent, right here, right now. The same is true of Gibsonian affordances, as can be illustrated with this quote from a textbook on ecological psychology (Michaels and Carello 1981).

Information is thereby depicted as to some degree 'personal,' as opposed to a detached list of qualities that could serve all organisms equally well. For example, a catalog of properties of a tree stump might include its shape and size, its texture, the wavelengths of light various parts reflect, moisture content or chemical composition in general, its position relative to other objects, and so on. But which of these properties permit a human to sit on it, or a termite to feed on it, or a bird to find bugs on it, or a rabbit to make a burrow under it? Qualities such as reflectance and size are certainly quantifiable, but they do not constitute useful, personal information. They are based on variables borrowed from traditional physics, and the only systems designed to detect such variables are instruments designed by physicists! There is no valid reason to suppose that evolution distributed this same roster of detectable qualities to animals with such vastly different informational needs.

Whether the stump affords sitting, eating, finding insects, or nesting-under depends on the animals effectivities [=abilities]. The stump does not afford nesting or eating to a human, nor does it afford sitting (on buttocks with vertical torso) to birds, termites and rabbits--their bodies cannot assume that anatomical configuration. Thus, the information that specifies affordances is personal to the animal that perceives it. (Michaels and Carello 1981, p.45)

The argument from this sort of understanding of the relationship between perception or thought and the world to a rejection of the Pyramid of Being is easy to imagine, but as we will see, it has not been sufficiently appreciated by all the proponents of strongly evolutionary theories of cognition. In what follows, I will spell out the argument that strongly evolutionary theories of cognition imply that the Pyramid of Being gets the natural world wrong.

## **2. Against the Pyramid of Being**

Here is the argument: Start with the assumption, common among proponents of strongly evolutionary theories of mind, that the most basic representations are action-oriented. As we

have seen, we should expect the so-called action-oriented representations of creatures to be about only those aspects of their environments that matter to the actions they regularly undertake. It follows from this that the world as represented by the action-oriented representations of animals with much different needs from those of humans will be much different from the world as represented by the action-oriented representations of humans. This is the case because throughout their evolutionary histories, animals develop perceptual systems responsive to opportunities to fulfill their needs. Because the needs of one type of animal can be so different from those of another, the perceptual systems that result will represent things very differently, a world full of barbecues and highways and myriad other things for humans, but, for example, a world containing only three things--what we see as butyric acid, pressure, and temperature changes--for ticks (See von Uexküll 1934, p.10).

Moreover, given the way evolution works, we should not think of the perceptual systems (or any parts of animals) as *ideal* solutions to problems posed by the environment.<sup>v</sup> Instead, animals that survive and reproduce are those that do well enough to find food and mating partners and so on. So, there is no reason to assume that any particular type of animal's perceptual system gets the world, as it is independently of thought, just exactly right; they all do only well enough.

Since there is no reason to except human perceptual systems, there is also no reason to think the world as humans perceive it has any special claim on being the true reality or world-in-itself from which ticks, poor things, know only three aspects. The everyday human perceptual world, far from being an objective, action-neutral environment, is just as "tightly geared to specific [human] needs and sensorimotor capacities" (Clark 1997, p.173) as the tick's perceptual world is geared to specific tick needs and sensorimotor capacities. Thus, the world as represented by the perceptual systems of humans (or any other type of animal) cannot be fully independent of those perceptual systems. If, as proponents of strongly evolutionary theories of cognition claim, the most basic representations are action-oriented (or pushmi-pullyu or affordances), it follows that cognitive agents, with their particular phylogenies and ontogenies, their physiology and education, play an important role in determining the nature of their perceived worlds.

In fact, there is good reason to think that perceptual worlds of different animals are incommensurable with one another. Many examples can be given to illustrate such a claim. One prominent among them is *color vision* (Varela, Thompson and Rosch 1991; Winograd and Flores 1986). Color vision, it is often claimed, shows the relativity of human experiences of color because different cultures have different color categories. This probably isn't true: humans, despite their differing languages and cultures, are able to register more or less the same eleven basic colors (Varela, Thompson and Rosch 1991, p. 168). The more interesting case for animals taking part in the construction of the world is made by considering differences *across species*. Human vision is trichromatic, our color vision can be represented along three dimensions (hue, saturation and brightness). Thompson, Palacios, and Varela (1992) compile evidence that other animals are dichromatic (rabbits), tetrachromatic (some birds) and perhaps even pentachromatic (some other birds). The color spaces of these animals need to be represented along two, four or five dimensions and so are *incommensurable* with ours. The colored world with which these animals are coupled is not the same one that we (humans) are coupled with; with their different

physiologies, birds and rabbits enact the colored world in a radically different way than we do. Thus the world to which these animals are coupled is not *the* world; it is *their* world.<sup>vi</sup>

So far proponents of strongly evolutionary theories of cognition will probably agree.<sup>vii</sup> But in so doing, they must also admit the falsehood of so-called "common-sense realism," in which the world-in-itself is thought to correspond to everyday human categories. For if humans and ticks and rats and so on are all able to live successfully in the very different worlds registered by their perceptual systems, in their very different *Umwelte* as von Uexküll (1934) would say, there can be no principled reason to privilege the human world (or any of the others) as the world-in-itself, as it really is independent of thought. Indeed cockroaches, for example, are arguably more successful than humans. Yet no one ever claims that we should adjust our ontology so that it meshes with the categories that cockroaches use. Thus, the ontology implied by our usual human categories has no special claim on representing the world-in-itself.

A common response to this argument is to claim that all of these categories are reflections of the world which is in-itself *physical*. The physical world, it might be said, is like an enormous cheese that can be sliced in many different ways. Different animals have evolved to slice the physical world into different sections, perhaps at different levels of granularity, but the categories of the successful animals will all be reflections of the underlying physical substrate.<sup>viii</sup> But it is a small additional step in the argument in the previous paragraph to see that physicalism or what Husserl (1970) called objectivism, in which the world-in-itself is taken to be the world as it is described by the mathematical idealizations of physics, is also false. Consider that Clark argues that "higher thought," the kind exhibited in mathematical and scientific theorizing, depends on the scaffolding provided by public language. Clark and Millikan, like Terrence Deacon (1997), also suggest that complicated mechanisms like those responsible for our language using abilities were adapted to the way our brains worked pre-linguistically; human language, they claim, is adapted to and built upon action-oriented representations (or pushmi-pullyu representations<sup>ix</sup> or affordances). As Millikan puts it:

We creatures who possess language have a long evolutionary history during which one after another of vastly many adaptive changes occurred. More sophisticated biological systems tend of course to develop not out of whole cloth but by modification of and superimposition upon more primitive systems. (1984, p.68; See also Clark 1997, pp.211-13)

As we have seen, our primitive representations are biased by pressures to fulfill human needs throughout human evolutionary history. And if the foundation on which language is built is biased, it is overwhelmingly likely that language itself is similarly biased. So if physics and other sciences depend upon our language-using abilities (and who can doubt that they do), they have no claim on being reflections of the world-in-itself. Indeed, the belief that the physical is somehow equivalent to the world-in-itself is especially baseless: as Carello and Michaels point out, nothing but the tools that physicists build are able to perceive things like reflectances and protons.

Supporters of what I have been calling strongly evolutionary theories of cognition (with the exception of Millikan, see below) resist these conclusions, and understandably so. Those

persuaded to accept an strongly evolutionary picture of the mind (as many do these days) cannot do as most philosophers of mind and cognitive scientists do; that is, they cannot just assume physicalism or common-sense realism.<sup>x</sup> This is especially consequential when trying to assign content to the mental states of non-human animals (not to mention states of artificially-evolved neural networks and controllers for animats). These states may frequently stand for objects and situations that have no clear analogue among either our common-sense or our scientific categories, as these categories do not necessarily reflect anything other than human needs and sensorimotor abilities.

Accepting this conclusion makes the task of cognitive science considerably more difficult: it changes from the (certainly challenging enough) investigation of the way agents process information about the objective environment, to an investigation of the what von Uexküll calls the *Umwelt* of agents being studied. Because of these consequences, we should expect objections to the suggestion that strongly evolutionary theories of cognition undermine the Pyramid of Being. In the next section, we will consider two such objections.

### 3. “Blocking the Exits”: On Two Possible Objections

There are two possible objections to the argument I have just made. One is that it gets evolution wrong; the other is that the argument makes stronger claims about the nature of the relation between cognitive system and world than are justified. In this section, I will address these objections. Doing so will clarify what I take to be the ontological consequences of the argument.

#### 3.1. Adaptationism<sup>xi</sup>

One possible objection to the argument above is that it is just false that evolution does not provide optimal solutions to problems posed by the environment. In fact, some might claim, that is precisely what evolution does. For example, *evolutionary epistemologists* like Popper (1972), Campbell (1974) and Lorenz (1962) have long assumed that the process of evolution guaranteed that our perceptual categories are accurate reflections of the world-in-itself, that our human *Umwelt* is guaranteed to be almost exactly like *Die Welt*. To make this claim is to be committed to a particularly strong form of what is known as *adaptationism*. If this strong form of adaptationism is correct, and evolution does optimize, the argument against the Pyramid of Being fails to go through. It fails because if the sensory systems of animals are optimal solutions to problems posed by the environment and scientific theories are extensions of our sensory systems, then physical science, which seems to be the most general scientific theory, will also be the most general theory of the world *as it really is*. In this section, I will characterize adaptationism briefly, then point out two common criticisms of it. The point will be to show that evolution does not optimize, and that it would be extremely surprising if the sensory systems of animals got the world just exactly right.

Adaptationism was initially described by Gould and Lewontin (in 1979) who wished to criticize it, as something of a caricature of the practice of actual evolutionary biologists. Gould and Lewontin (1979) claim that adaptationist theorizing generally proceeds in two steps. First, "[a]n organism is organized into 'traits' and these traits are explained as structures optimally designed by natural selection for their functions." (1979, p.77) And second,

After the failure of part-by-part optimization, interaction is acknowledged via the

dictum that an organism cannot optimize each part without imposing expenses on others. The notion of trade-off is introduced and organisms are interpreted as best compromises among competing demands. Thus interaction among parts is retained completely within the adaptationist program. And suboptimality of a part is explained as its contribution to the best possible design for the whole. (1979, p.75)

Surprisingly, Dennett, in his book length defense of adaptationism (1995), endorses a very similar definition of it, proposed later by Lewontin.

[Adaptationism] is defined by one of its most eminent critics as the "growing tendency in evolutionary biology to reconstruct or predict evolutionary events by *assuming* that all characters are established in evolution by direct natural selection of the most adapted state, that is, the state that is the optimal 'solution' to a 'problem' posed by the environment." (Lewontin 1983) These critics claim that, although adaptationism plays *some* important role in biology, it is not really all that central or ubiquitous--and, indeed, we should try to balance it with other ways of thinking. I have been showing, however, that it plays a crucial role in the analysis of every biological event at every level from the first self-replicating macromolecule on up. (Dennett, 1995, p.238)

Given that both the most eminent critics and the most ardent defender of adaptationism accept the same definition, we will also accept it. Thus, adaptationism is a methodology whereby one treats an organism as a set of traits, each of which is selected for because it is the optimal structure available to selection for the fulfillment of its function.

The first line of criticism of adaptationism concerns the assumption made by adaptationist evolutionary biologists that natural selection is guaranteed to *optimize*, that is, to produce a feature of an organism that is ideally suited to perform some function made necessary by the organism's environment. It turns out that this assumption is simply false. Kitcher (1985) points out several ways in which natural selection will fail to yield an optimal solution to the "problem" posed by the environment. First, he describes the course of evolution at a locus in a Bantu population. The combination of alleles available in this population and their relative fitnesses combine so as to lead to the demise of the fittest combination of alleles.<sup>xii</sup> (See Kitcher 1985, p. 215-6) Furthermore, there are several other ways in which the most fit combination of alleles just cannot be reached by the population. Kitcher calls this the *trajectory problem* ("Intuitively, the idea is that you can't get there from here." (1985, p.216)) Such trajectory problems can be caused by a variety of factors such as small populations, various vagaries of chance, and interactions among gene loci. When there are trajectory problems, the best "solution" that can be hoped for is a *local* optimum, which is best described by altering Kitcher's description of the trajectory problem--intuitively, the local optimum is "the best you *can* get to from here". And it is often the case that once the population settles into this local optimum, the global optimum is no longer an available option. Thus, the adaptationist is wrong in assuming that natural selection is *guaranteed* to optimize.

The other line of criticism of adaptationism that is relevant here stems from what is sometimes called *free riders*, cases where selection *of* objects, rather than selection *for*, certain traits has occurred. Sober (1984) has characterized this distinction in an illuminating way. He

asks us to imagine a toy with balls of different colors and sizes (which are correlated in that every ball of a certain size is the same color) inside a cylinder that is divided into several chambers. Each divider between two chambers has holes of a particular size, with the largest holes between the top and second chamber, the next largest holes between the second third and so on all the way down (see Sober, 1984, p.99 for a picture of this toy). With such a toy, if all the balls are placed in the top chamber and the toy is shaken, the balls will eventually descend to the lowest chamber with holes above it that they can fit through. Suppose, as Sober does, that only the green balls are small enough to fit through all the holes and reach the bottom chamber.

When the green balls reach the bottom more frequently than the blue ones, we think that there must have been a reason why. Green balls were selected; so they must have had some property that was selected for. But the property in question was not their color. There was *selection of* green objects, but no *selection for* greenness. I offer the following slogan to summarize this logical point: "selection of" does not imply "selection for". (Sober, 1984, p.100)

What this example shows is that since it is only properties that can be *selected for* and any object has many properties, there are likely to be evolutionary *free riders*, properties of objects of which *selection of* occurred, but without *selection for* the particular property in question. The greenness of the balls that were selected is just such a case--there was selection of the green objects, but this selection was selection for size, not greenness.

One variety of free rider is what Gould and Vrba (1982) call *exaptations*. Exaptations, they say, are characteristics that "evolved for other purposes (or for no function at all), and [are] later 'coopted' for their current role." (1982, p.6) So exaptations begin as free riders, and only later become useful for some purpose or other. Adaptationism is incorrect, they say, because it fails to consider that certain traits might be exaptations. To return to Sober's toy example, imagine that the greenness of the balls, which, remember, came along as a free rider when the smallness was selected for, suddenly becomes quite a useful trait. Imagine, for example, that the balls are seeds and birds in their environment are, for whatever reason, particularly good at spotting green seeds, which they then ingest and excrete at some distant points, thereby spreading these green seeds, and the plants they grow into, widely. There was *selection for* the smallness of the seeds, and *selection of* the seeds themselves, including such traits as their greenness. Thus, Gould and Vrba would say, the smallness of a seed is an adaptation, the greenness is an exaptation. Since adaptations have functions and exaptations only have *effects*, the greenness of the seeds has no function.<sup>xiii</sup> Gould and Vrba even suggest that the cognitive and language-using abilities of humans may be underwritten by exaptations in a brain whose features were selected for quite different reasons. ("Most of what the brain now does to enhance our survival lies in the domain of exaptation--and does not allow us to make hypotheses about the selective path of human history." (1982, p.13)) This, then, is a direct assault not just on the claim that evolution does not optimize, but also on the very plausibility of strongly evolutionary theories of cognition. For if the features of brains that underlie our representation using abilities are exaptations, if they exist in current humans as free riders that came along with some other trait that was selected for, not only are they unlikely to be optimal reflections of the world-in-itself, but they will have no functions. And for states of animals to have content according to

strongly evolutionary theories of cognition, they must have functions.

Millikan (1993) deals with Gould and Vrba's critique by expanding her definition of function so that free riders have functions. She says:

Being "built" by natural selection is sufficient for proper function, being maintained by natural selection is independently sufficient, and having been utilized by other structures built or maintained by natural selection is also independently sufficient for proper function. It is hard to see how modern human cognition could fail to be caught in one or more of these nets. (1993, p.49)

These last two criteria for having proper functions are designed precisely to deal with these critiques of adaptationism. The first of these additional criteria, being *maintained* by natural selection, makes it such that exaptations have functions. This is the case because all current adaptations are (or once were) exaptations--they only *begin* as free riders. A thought experiment from Dennett 1995 makes this point vividly. Imagine that some American engineers invent a device to recognize U.S. Treasury twenty-five cent pieces in American vending machines. Call such a device a "two-bitser" and imagine that it is (re)produced and placed in most vending machines. Each "two-bitser," then, is a member of a reproductively established family and was selected *for* (by manufacturers of vending machines) because of its ability to recognize quarters, so its function is to do just that. Now imagine that after years of working in the United States, a vending machine that contains a two-bitser is brought to Panama, where there is a coin called (this is true, Dennett tells us) a "quarter Balboa" that is the same shape and weight and approximate value as U.S. quarters. The two-bitser, it turns out, is also an excellent detector of quarter-balboas, and is used as a "q-balber" for several years. It is then an *exaptation* for the detection of quarter balboas. So, Gould and Vrba would say, it has as an *effect* the detection of quarter balboas, but its *function* is to detect U.S. quarters. (see Dennett, 1995, 404-407) But now, going a small step beyond Dennett's thought experiment, imagine that a Panamanian manufacturer decides to build lots more of these devices (*maintaining* them by selection) precisely because of their ability to detect quarter balboas. The device now meets the conditions of having acting as a "q-balber" as its function.

What this thought experiment is intended to show is that adaptations must all have been features of an organism that were already present for some other (or no particular) reason, and were then pressed into service in a certain job. In other words, all adaptations must have begun as exaptations. So either exaptations are very short-lived as exaptations (because once they are reproduced *because* they can perform the effect they were exapted for, they become adaptations, and so are no longer exaptations) or exaptations have functions (because adaptations do and all exaptations are also adaptations).

Though Dennett's thought experiment is vivid, there is one way in which it is unrealistic. Because quarter Balboas were made in American mints from the same stock as U.S. quarters, the two types of coins are very nearly identical. So the two-bitser will work almost perfectly when pressed into service as a q-balber. But most cases of exaptations will not work so well; they are much more likely to be just serviceable, gerrymandered, and stopgap. The following story about Mongolia is a more realistic, non-biological example of an exaptation. Mongolians generally have very little interest in being converted to Christianity by American missionaries. But

missionaries are generally welcome guests because they pass out cheap copies of the *New Testament*. These books were designed to educate people about the teachings of Jesus, make converts of their readers, provide inspiration; most generally, they were designed to be read. Mongolians seek them out, however, for another purpose entirely: the pages, despite the ink, make passable rolling paper for cigarettes--something too expensive for most Mongolians to afford. The paper does not work perfectly: it burns too fast and the ink is potentially more carcinogenic than tobacco, but it works well enough that the *New Testaments* are kept around (*maintained*), even sought out, as rolling paper. They have been exapted for the purpose of being smoked, and, by the first of Millikan's new criteria (the second criteria overall), that is their function.

Millikan's second new criterion (third overall) by which something can be a function, having been utilized by other structures built or maintained by natural selection, is intended to allow things that have been exapted by the construction of new surrounding mechanisms to have functions. The example that Millikan discusses is the flippers of sea turtles. These flippers are clearly designed for swimming, but are also serviceable as digging implements when the turtles come on land to lay eggs each year. What makes exaptations such as this have functions is that there are other parts of the sea turtle that have been selected for their abilities to work with the flippers to enable them to be used for digging; in particular, brain mechanisms have evolved for the purpose of digging with the flippers (Millikan 1993, p.45). Similarly, if one builds a go-cart body, a steering mechanism, and gear mechanism around an old lawn mower motor, it is the function of the motor to propel the go-cart at (what my parents claimed were) dangerous speeds. Also similarly, if one follows any of the recipes in *The Anarchist's Cookbook* to make incendiary devices from household items.

These replies to Gould and Vrba guarantee that exaptations do have functions. Indeed, unless one is willing to countenance some sort of *pre-selection*, in which the evolutionary process is credited with the foresight to preserve traits that will someday be useful for some purposes despite the fact that it is not useful now, *everything* that is an adaptation will have once been an exaptation. With Millikan's second new criterion, we can see that many adaptations will be like the sea turtle's flippers or a bomb built from a light bulb and gun powder--items that have functions in virtue of having been preserved more or less as they were but with Rube Goldberg devices built-up around them to make them suitable for some other purpose.

In general, these exaptations will be far from optimal solutions to problems posed by the environment. Assuming sensory systems are built this way as well, they will be gerrymandered, just good enough, satisficing solutions. It would be just by luck if exaptation-based adaptations were accurate reflections of the world that exists independently of them.

### 3.2. "Many Slicings, One Cheese"

A second possible objection to the argument offered above is one made by Barry Smith (unpublished). In response to an argument by Katz (1987), Barry Smith writes the following:

But, to remain with Katz's own preferred example, space (as we may here assume) is a continuum. Like all continua it can be partitioned in a range of mutually incompatible ways (as a cheese can be sliced in such a way as to produce either triangular or rectangular or disk-shaped segments, but not all of these at once). All

members of a family of mutually conflicting ‘perceptual spaces’, now, may very well turn out to be compatible after all, if they are expressing distinct *partitions*, for example partitions on different levels of granularity, of one and the same reality. The world (like a lump of cheese) can be sliced into pieces in more than one way. (unpublished manuscript, p.13)

Lorenz’s *evolutionary Kantianism* (1962) is another example of such a position. Lorenz argues that evolution has built something like Kant’s *a priori* into the sensory systems of animals. But unlike Kant, Lorenz supposes that we can know something of the world-in-itself, because the world has selected sensory systems (and their innate, structuring *a prioris*) for their fit to the world-in-itself. Should this be the case, the argument I offer in above will fail to have the consequences that I claim that it does. It will indeed be the case that the *Umwelte* of particular types of animal will not be fully independent of those animals (and their evolutionary history), but this will have no metaphysical consequences because their *Umwelte* are groups of parts or aspects of the world-in-itself. All that will depend on the organism is that the particular partition or “slicing” of the world-in-itself, which existed independently, is currently being actively used by some type of animal.

The question to ask someone who holds a position like Lorenz’s is: What is the nature of the world-in-itself? Evolutionary epistemologists have typically assumed that the world-in-itself is the world as described by *physics*. As Campbell puts it, after explicitly rejecting common sense realism, in which the world-in-itself is the human perceived world:

The different *Umwelte* of different animals do represent in part the differential utilities of their specific ecological niches, as well as differential limitations. But each of the separate contours diagnosed in the *Umwelte* are also diagnosable by a complete physics, which in addition provides many differentia unused and unperceived by any organism. (1974, p.448; see also Lorenz 1962 and Popper 1972.)

On such an understanding, the sensory systems of animals represent the physical world, whether or not the animal itself has any understanding of the world *qua* physical. They will represent the physical world because it is the physical world that has selected the sensory systems for their abilities. But we have seen (in the argument in section 2) that there is no reason for having such confidence in physics, there is no reason to assume it will play that role.

Once one drops the assumptions of optimizing adaptationism and physicalism, as I have argued one should, the logic of the evolutionary process is changed.<sup>xiv</sup> We can no longer think of evolution as providing optimal solutions to problems posed by the (physical) world-in-itself. Varela, Thompson and Rosch (1991) have put it this way:

The first step is to switch from a prescriptive logic to a proscriptive one, that is, from the idea that what is not allowed is forbidden to the idea that what is not forbidden is allowed. In the context of evolution this shift means we remove selection as a prescriptive process that guides and instructs in the task of improving fitness. In contrast, in a proscriptive context natural selection can be seen to operate, but in a modified sense: selection discards what is not compatible with survival and reproduction. Organisms and the population offer variety; natural selection guarantees only that what ensues satisfies the two basic constraints of survival and

reproduction. (1991, p.195)

Evolution, then, rather than actively building organisms that are well-adapted to the world, eliminates those that are too ill-suited for survival. It may have no effect on those parts of organisms that are not actively detrimental to survival, allowing the continuance of many traits that are merely neutral with respect to survival and reproduction. As Varela, Thompson and Rosch put it, “much of what an organism looks like and is ‘about’ is completely *underdetermined* by the constraints of survival and reproduction.”(1991, p.196)<sup>xv</sup> The case of color vision discussed above is a good illustration of this claim. The color spaces of many non-human animals are *incommensurable* with ours. The colored world with which these animals are coupled is not, indeed given the incommensurability *cannot be*, the same one that we (humans) are coupled with. But notice that the evolution allows all of these different colored worlds, despite their incommensurability.

The role of the world-in-itself in such a picture of evolution is greatly diminished: all viable *Umwelte* are allowed, not just those that are well-adapted to or accurate reflections of the world-in-itself. These viable *Umwelte* will not be independent of the animals that enact them. The relationship between the *Umwelt* and the animal will be what Barry Smith (unpublished) has called *generic dependence*. He defines it as follows: x is generically dependent upon objects of sort S if and only if x is necessarily such that it cannot exist unless some object of sort S exists. So the set of affordances that makes up the snake’s world will depend upon there being at least one snake; as long there are snakes, there will be snake affordances. There is, of course, still a role for the world-in-itself. Although there is no reason to assume that the *Umwelte* of animals (or parts of them) will correspond to the world-in-itself (or parts of it), each of these *Umwelte* must be such that it does not go so far astray from the world-in-itself that the continued existence of the type of animal is compromised. But note that the *Umwelte* of particular types of animals will depend *specifically* on the world-in-itself: if there were no world-in-itself, there would be no *Umwelte*.

The world-in-itself here plays what Harré and Krausz (1996) call a *regulative* role; that is, an independent world-in-itself is a necessary posit for strongly evolutionary theories of mind (as well as natural science) to make sense. The *Umwelte*, on the other hand, play *criteriological* roles; the mental states of animals will represent entities in the *Umwelte* of those animals.

#### 4. Conclusion

I have argued that maintaining a strongly evolutionary theory of cognition forces one to reject the conception of nature in which the world is taken to be *at bottom* physical. In so doing, one need not reject naturalism or embrace any kind of dualism. The point instead is to refuse to grant pride of place to the physical, to take it off the bottom of the Pyramid of Being (figure 1). Indeed, the physical should be seen as a late and decidedly nonbasic way of being, built as it is upon human affordances and language, which are themselves only a satisficing representation of the world. According to the metaphysical picture that follows from strongly evolutionary theories of cognition, which we might call *Biological Pluralism*, we should believe in the entities of any set of categories that the world supports.<sup>xvi</sup> Each of them is a *good enough* reflection of what the world-in-itself is like to support the survival of a species of animals, and has been

refined over millenia by evolution. Of course, none of them, including our own, will be perfect. So there is no reason to assume that the world as constituted by human affordances is in any way “more real” than the world as constituted by the affordances of any other variety of animal. What is true of human affordances will also be true of human sciences, as they depend upon human perceptual systems.

It is important to note that, as I mentioned above, biological pluralism is in no way non-naturalistic. Indeed, it can be seen as a version of naturalist *extraordinaire* Ruth Millikan’s ontological picture.<sup>xvii</sup> Perhaps alone among the proponents of strongly evolutionary theories of cognition, Millikan has appreciated that her position has metaphysical consequences. To see that Millikan’s position should be seen as biological pluralism, consider that she distinguishes between principles of *identity*, which concern how the world actually is, and principles of *unity*, which concern how sensory systems or languages divide the world into perceptual and linguistic categories. She writes

The principle of strict identity over time is not strong enough by itself to divide up the world into a determinate set of enduring objects. Or more accurately, it divides the ongoing world into chunks that are far too numerous, overlapping, and crisscrossing to try to keep track of. So a great deal of room is left for decision on our part as to how to divide the world into sensible-sized and sensibly unified temporally extended wholes to be recognized and honored by coordination with common and/or proper names. (1984, p. 293)

Thus, evolution constructs animals with sensory systems (and humans with language-using abilities should these be innate) that obey some principles of unity, and these principles of unity must reflect world’s (N.B.: *not* the world as described by physics) principles of identity, but there are many, many ways that principles of unity can differ and still not run afoul of worldly identities. So, human common sense, human language, human science and cockroaches will all be built to honor principles of unity, each of which will be different ways of respecting the identities of the world.<sup>xviii</sup>

As I pointed out above, there are significant costs to accepting these consequences. For one, cognitive science becomes more difficult. Rather than studying the way that animals interact with the entities in our common-sense world or the world as described by physics, one must investigate their *Umwelte*, which in many cases will be very different from our own. Strongly evolutionary cognitive science, it seems, must combine ethology, neuroscience, evolutionary biology, with elements more traditional cognitive science to discover the nature of the *Umwelte* of non-human animals and artificially evolved robots. Understanding these presumably simpler *Umwelte* will provide clues to the nature of the human *Umwelt*. Such work will be much different from business-as-usual cognitive science, but there are some model cases of it that have been done. Kathleen Akins’s “What is it like to be Boring and Myopic?” (1993), “Ways of Coloring” by Thompson, Palacios and Varela (1992), and “Evolutionary Robotics: The Sussex Approach” by Harvey, Husbands, Cliff, Thompson and Jakobi (1997) are prime examples.

But more importantly than this change in the *methodology* of cognitive science, cognitive scientists and philosophers of mind must now take up a more traditional metaphysical project:

explaining *objectivity*. Once one denies that the Pyramid of Being is an accurate depiction of nature, one must work to explain how the mind is and how the world is, in such a way that accounts for the possibility of using language and doing science in the way we do. That is, since it is assumed that the relation between mind and world is at least partly constructive, it must be explained how it is that our linguistic categories take hold, and how some of our theories and sentences are *true*. This task (which has been begun by Brian Cantwell Smith 1996, Haugeland 1997 and Cussins 1990) is admittedly a much more daunting one than that facing the traditional cognitive scientists. But it is one that we must be committed to.

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<sup>i</sup>This duality between supervenience and implementation is not often noticed. Brian Cantwell Smith pointed it out to me. See Smith forthcoming.

<sup>ii</sup>But I will not defend strongly evolutionary theories of cognition here. I have done that elsewhere (Chemero 1998a, Chapter 1).

<sup>iii</sup>An immediate consequence of this is that Papineau's (1993) conception of naturalism, in which naturalism *just is* physicalism, cannot be right.

<sup>iv</sup>Similar assumptions are made by many robotocists. See, for example, Beer 1995 and Brooks 1991.

<sup>v</sup>This is not to say that evolution is not understandable as reverse engineering, as Dennett (1995) has recommended. The claim is rather that Mother Nature prefers quick and dirty solutions over those that get things absolutely right. Even Dennett, the most vocal proponent of adaptationist biology, should agree with this. For more on adaptationism, and its relation to this argument, see section 3.1 below.

<sup>vi</sup>The incommensurability of these perceived worlds makes the objections to an earlier version of this argument raised by Mandik and Clark (unpublished) moot. They claimed that this argument goes wrong in two ways. First, they say that the argument moves from "X represents only Y and Z" to "X represents the world as containing only Y and Z". The latter is incompatible with objectivism, the former is not. But if different types of animals represent incommensurable worlds, then we have "X represents Y, where Y implies not-Z" and "W represents Z".

Second, they claim that the argument ignores the possibility that it could be the function of some animal's representational system to represent both things in the animal's *Umwelt*, and also things in the world-in-itself. But this is impossible if the *Umwelte* of different animals are incommensurable.

<sup>vii</sup>In *Kinds of Minds* (1996), Dennett writes:

...we tend to forget that *our* ways of thinking about the world are not the only ways, and in particular are not prerequisites to engaging the world successfully. It probably seems obvious, at first, that since they are so manifestly intelligent, dogs and dolphins and bats must have concepts more or less like ours, but on reflection it shouldn't seem obvious at all. Most of the questions we've raised from our evolutionary perspective about the ontology and epistemology of other creatures have not yet been answered, and the answers will no doubt be surprising. (1996, p.146)

<sup>viii</sup>The cheese analogy is from Barry Smith's "Social Objects" (unpublished), but similar ways of thinking are evident in Gibson 1979, Michaels and Carello 1981, and Clark 1997. Barry Smith, it is worth noting, has tried to formalize the ontology that results from the multiply-sliced-cheese understanding of the physical world. The ontological picture that results is radically different from traditional realist ontology; in particular, Smith rejects set theory as an ontological tool, and switches to what he calls 'mereotopology'. See Smith unpublished. For more on this see below.

<sup>ix</sup>Millikan agrees in an unpublished paper “Some Different Ways to Think” (Millikan unpublished).

<sup>x</sup>To his credit, Clark realizes that his picture mind leads to a push away from physicalism, or objectivism (Husserl 1970). Unfortunately, he dismisses the point extremely lightly. In his discussion of the relationship between his project and a similar one described in Varela, Thompson and Rosch (1991) which actually does draw conclusions opposed to physicalism and objectivism, Clark says the following:

...Varela et al. use their reflections as evidence against realist and objectivist views of the world. I deliberately avoid this extension, which runs the risk of obscuring the scientific value of an embodied, embedded approach by linking it to the problematic idea that objects are not independent of the mind. My claim, in contrast, is simply that the aspects of real-world structure which biological brains represent will often be tightly geared to specific needs and sensorimotor capacities. (1997, p.173)

<sup>xi</sup> This section draws on Chemero (1998b)

<sup>xii</sup>The alleles and fitnesses are as follows: A, S and C are initially present; most members of the population are AA; AS is fitter than AA; SS is lethal; C is recessive to A; CS is less fit than AA; CC is the most fit pair. In such a population, C is eliminated, and so the most fit pair CC is also eliminated, despite the fact that it was present initially. (See Kitcher 1985, p.215)

<sup>xiii</sup>The (now classic) example Gould and Vrba discuss is the flippers of sea turtles. The flippers, they claim, are adaptations for swimming that also serve reasonably well as diggers, an activity for which they have been exapted. See Millikan 1993, pp. 41-50, for a criticism of this particular case.

<sup>xiv</sup>It is still possible to maintain either of adaptationism or the claim that the world-in-itself is the physical world. The only reason to do so, in light of the arguments I have outlined against them, would be a prejudice in favor of the pyramid of being. Many cognitive scientists and philosophers of mind share this prejudice, but it is *only* a prejudice.

<sup>xv</sup>This view of evolution is instantiated by work on robotics at Sussex. The genetic algorithm called SAGA used there does not choose the best performers at some task to be parents in the subsequent generation. Instead, all members of the population whose *worst* trials were successful become parents. See Harvey et al 1997.

<sup>xvi</sup>Note that this pluralism differs from that of van Gelder 1998. van Gelder argues that minds are ontologically complex, so the world must contain many different ontological types. I am arguing that there must be many different kinds of worlds.

<sup>xvii</sup>Dennett for one, has claimed that compared to Millikan, “virtually everyone else writing about meaning, reference, and intentionality [...] can be seen as failed naturalists.”(Dennett 1984, p.ix)

<sup>xviii</sup>Note also that Millikan’s main ontological category in her theory is what she calls a *substance*, and that she claims that substance is “at root an epistemological category”(1984,

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p.275).